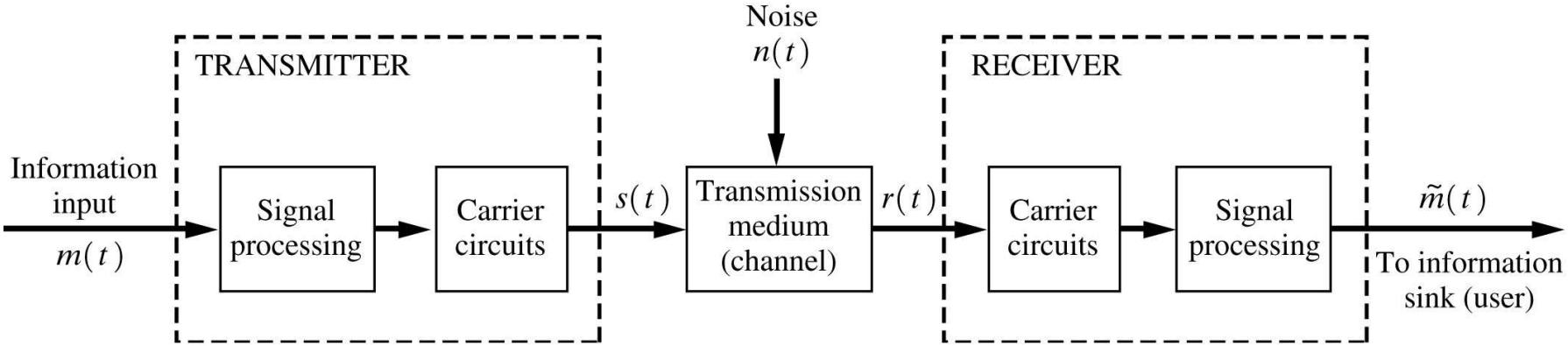
Communication Systems Lecture-1: Introduction

Chadi Abou-Rjeily

Department of Electrical and Computer Engineering
Lebanese American University
chadi.abourjeily@lau.edu.lb

August 27, 2024

Block Diagram of Communication Systems (1)



All communication systems involve three main subsystems:

- Transmitter
- Channel
- Receiver

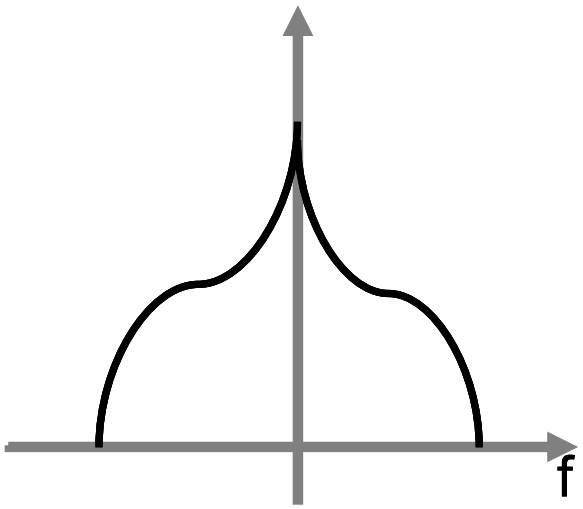
Block Diagram of Communication Systems (2)

- The message from the source is represented by the information input waveform $m(t)$.
  - This message can be audio, video or any other type of information.
- The message delivered to the receiver is denoted by $\tilde{m}(t)$.
- At the output of the communication channel, the received signal is:
  - Attenuated
  - Distorted: due to undesired filtering or undesired nonlinear effects in the system.
  - Corrupted by noise $n(t)$.
- Consequently, the received message may not be always the same as the transmitted message:

$$\tilde{m}(t) \neq m(t)$$

Block Diagram of Communication Systems (3)

- The Signal-Processing block at the transmitter conditions the source for more efficient transmission.
- The message $m(t)$ is a **baseband signal**:
  - It has frequencies located near $f = 0$.
  - Audio signals: Humans hear frequencies between 100 Hz and 3 kHz.
- The signal at the output of the signal processor is also a baseband signal.

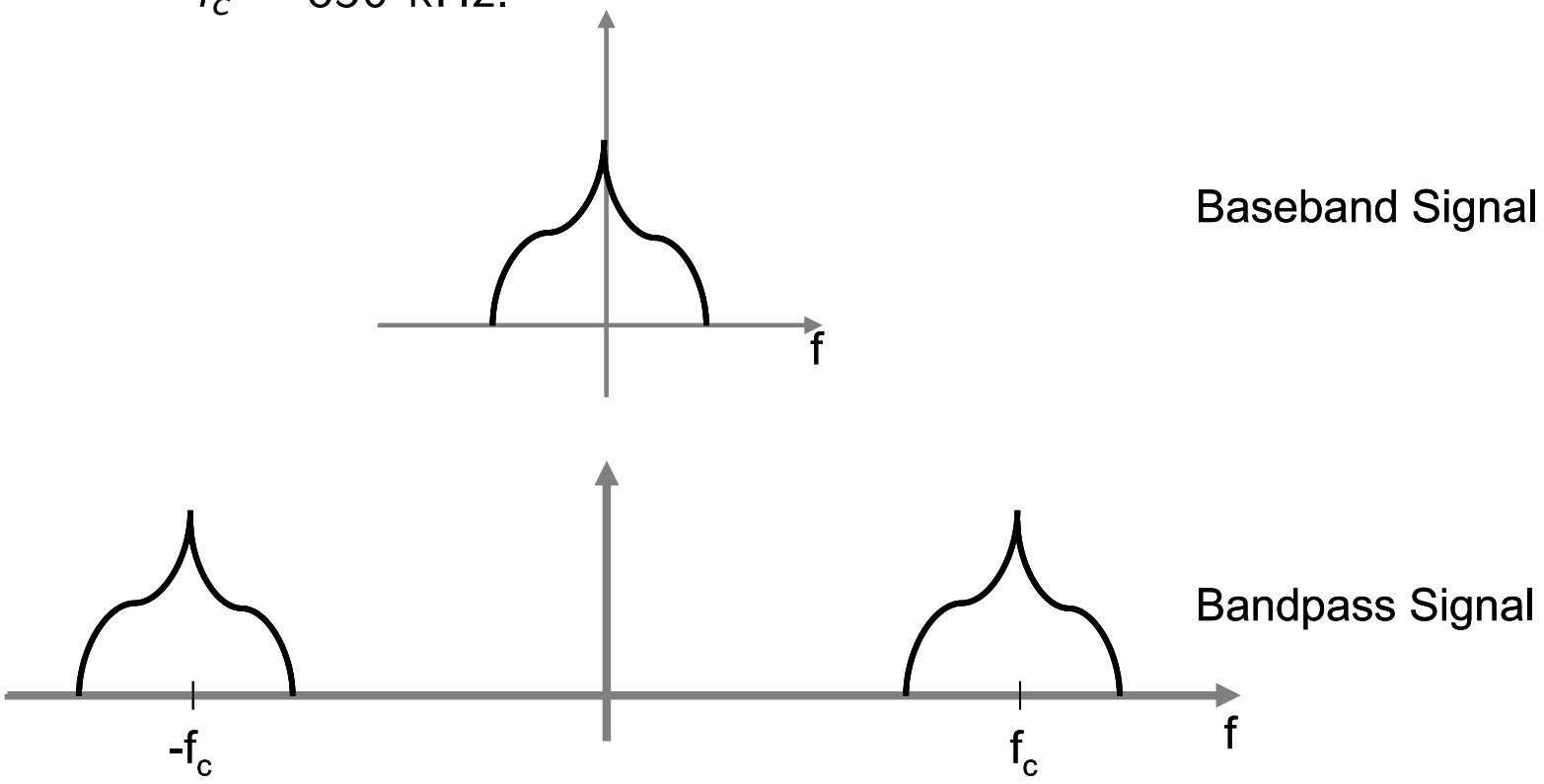


Block Diagram of Communication Systems (4)

- Channels may be classified as:
  - Wired Channels:
    - Twisted-pair telephone lines.
    - Coaxial cables.
    - Waveguides.
    - Optical fibers.
  - Wireless Channels:
    - Air: radio communications.
    - Water: under-water communication systems.
- Baseband signals can not propagate over most of the channels. Consequently, carrier circuits are needed at the transmitter and the receiver.
- The transmitter carrier circuit converts the processed baseband signal into a frequency band that is appropriate for the channel.
- If the channel propagates baseband signals, no carrier circuits are needed.

Block Diagram of Communication Systems (5)

- The mapping of the baseband signal into a bandpass signal $s(t)$ is called modulation.
- The frequency components of the modulated bandpass signal are located around the carrier frequency f_c.
  - An amplitude-modulated (AM) broadcast radio station with an assigned frequency of 850 kHz has a carrier frequency of $f_c = 850$ kHz.



Block Diagram of Communication Systems (6)

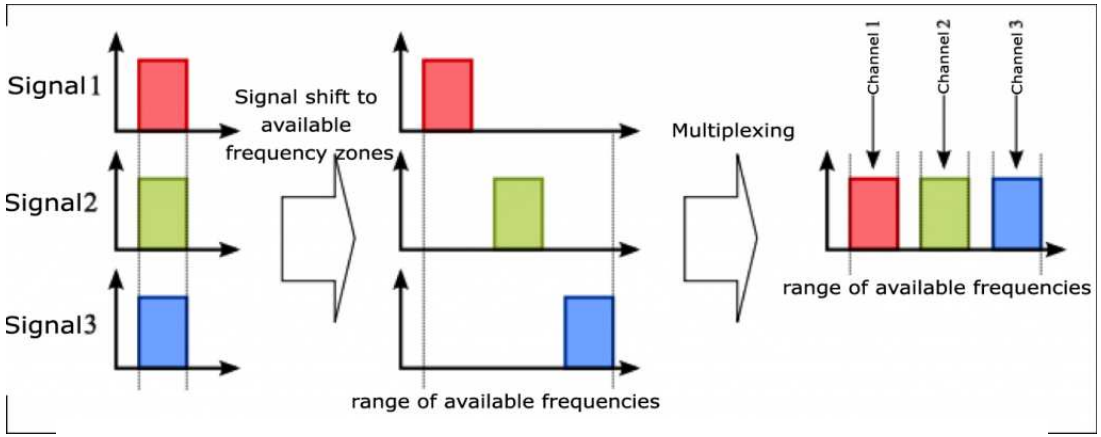
- Example-1: Optical fiber communications:
  - The carrier circuit converts the baseband input to light frequencies and the transmitted signal $s(t)$ is light.
- Example-2: Wireless communications:
  - The size of the antenna depends on the wavelength of the transmitted signal (for example, $l = \lambda/2$).
  - Consider an audio signal having a maximum frequency of 3 kHz. The minimum wavelength of this signal is:

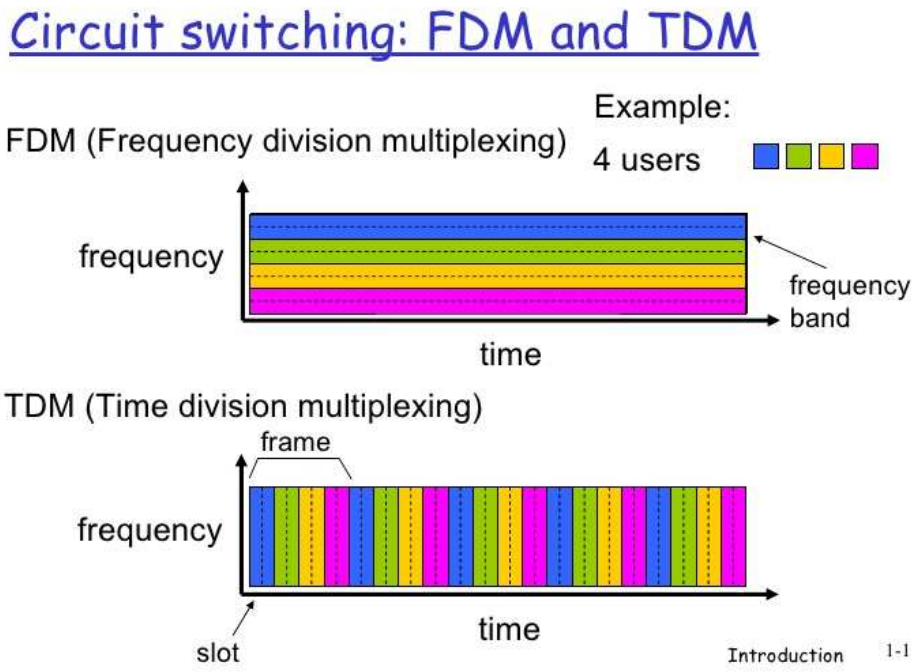
$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \quad (\mathrm{m/s})}{3 \times 10^3 \quad (1/\mathrm{s})} = 100 \quad \mathrm{km}$$

  - Consequently, baseband signalling is not possible in this case since it necessitates an antenna of 50 km.

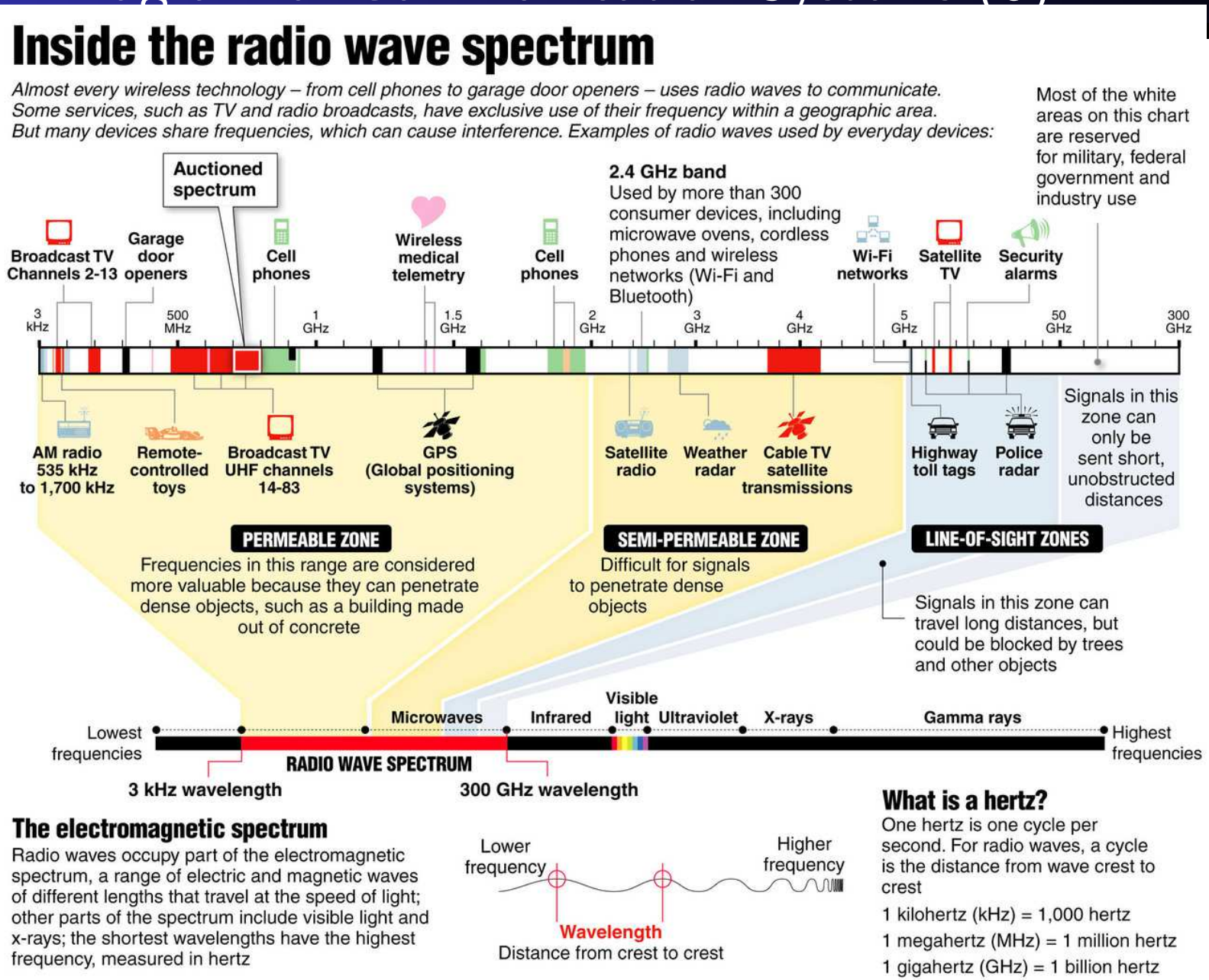
Block Diagram of Communication Systems (7)

- Frequency Division Multiplexing:





Block Diagram of Communication Systems (8)



Block Diagram of Communication Systems (9)

It can be shown that any passband signal can be written as:

$$s(t) = R(t)\cos\left[\omega_c t + \theta(t)\right]$$

where:

- $\omega_c = 2\pi f_c$.
- The angular frequency is equal to $\omega_c + \theta'(t)$.
- The modulation process causes $R(t)$, $\theta(t)$ or both to change as a function of $m(t)$.
- The information is carried by the amplitude, phase and frequency of the analog signal $s(t)$.
  - Example: for amplitude-modulation (AM), $\theta(t) = \text{constant}$ while the message $m(t)$ is contained in the amplitude of the signal ($R(t)$).

Design of a communication system

The goal is to design communication systems that transmit information to the receiver while:

- Introducing the least possible distortions to the information messages (quality).
- Transmitting as fast as possible (quantity).

The above objectives must be satisfied while verifying the following design constraints:

- The allowable transmitted energy.
- The allowable signal bandwidth.
- Cost.

Objectives of the course

- Baseband Signalling:
  - Transceiver design.
  - Different modulation schemes: Pulse Amplitude Modulation (PAM), Pulse Code Modulation (PCM), Pulse Width Modulation (PWM), Pulse Position Modulation (PPM) ...
  - Performance.
    - Signal-to-noise ratio.
    - Bandwidth.
  - Complexity.
- Bandpass Signalling:
  - Transceiver design.
  - Different modulation schemes: Amplitude Modulation (AM), Phase Modulation (PM), Frequency Modulation (FM), Phase Shift Keying (PSK), Frequency Shift Keying (FSK) ...
  - Performance.
    - Signal-to-noise ratio.
    - Bandwidth.
  - Complexity.